

9. Team Reflection Template

Submit this 1-page reflection as part of your Phase 4 deliverable. Be SPECIFIC — generic answers ('teamwork was hard') are not useful.

Project #: ____

Project Title: ____ self balancing robot with 2 DOF ____

Category: A / B / C / D / E (circle one)

Team Members: ____ chaitanya h93, apurv h95 ____

Date: ____ 24/4/26 ____

Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The hardest part was getting the mathematical model for the balancing part (inverted pendulum) correct, especially relating the tilt angle with the movement of the wheels. Initially, it was confusing to combine the rotational motion and linear motion into proper equations.

I resolved it by breaking the system into smaller parts—first understanding the pendulum separately and then the wheel motion—and then combining them step by step. Referring to standard inverted pendulum models and simplifying assumptions (like small angle approximation) also helped make the equations manageable.

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

CO3 was the most challenging for our team, as it involved applying the theoretical concepts to a practical system. While we understood the theory in lectures, actually using it to model and analyze a real system like the self-balancing robot was difficult. It required a clear understanding of multiple concepts together and careful implementation, which took time and a few iterations to get right.

Q3: What would you do differently if you started the project over?

If we started the project again, we would spend more time in the initial planning and modeling stage. In our case, we jumped a bit quickly into implementation, which led to some confusion later. We would also try to test each part of the system separately (like sensors and motors) before integrating everything, so debugging becomes easier. Overall, a more structured approach from the beginning would save time later.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

This project taught us how theoretical concepts actually behave in real systems. In textbooks, problems are ideal and straightforward, but here we had to deal with things like sensor noise, delays, and tuning the system through trial and error. We also learned how different parts—mechanical, electrical, and control—work together, which is not something you fully understand from solving only written problems.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

We divided the work between the two of us based on our strengths. One of us focused more on the mathematical modeling and derivation part, while the other handled coding and hardware implementation. However, we regularly discussed our

progress and explained our parts to each other. So even though we worked on different sections, both of us understood the overall derivation and how the system works.

Q6: What AI tools did you use? For what specific purpose? What did you NOT use AI for?

We used AI tools mainly to clarify concepts and understand the theory better, especially for topics like the inverted pendulum model and PID control. It also helped in checking equations and getting a general idea of how to approach the problem.

However, we did not rely on AI for the complete derivation or final answers. The actual modeling, calculations, and implementation were done by us, as we needed to understand the process ourselves for the project.